Arrays & Strings — 20 Practice Problems

Beginner Problem Set with Descriptions, Test Cases & Explanations

Part 1: Basic Array Problems (Questions 1 – 6)

Problem 1: Find Maximum and Minimum Element in an Array

Very Easy

Description: Given an integer array `arr` of size N, write a program to find and return the largest (maximum) and smallest (minimum) numbers present in the array.

Sample	Input Array	Expected Output
Test Case 1	`[3, 5, 1, 9, 2]`	`Max: 9, Min: 1`
Test Case 2	`[-10, 0, 45, 12, -3]`	`Max: 45, Min: -10`

Explanation (TC 1): Scanning through [3, 5, 1, 9, 2], 9 is the highest value and 1 is the lowest value.

Problem 2: Reverse an Array In-Place

Very Easy

Description: Given an array `arr`, reverse its elements so that the first element becomes the last, the second becomes second-last, and so on. Try doing it in-place using two pointers.

Sample	Input Array	Expected Output
Test Case 1	`[1, 2, 3, 4, 5]`	`[5, 4, 3, 2, 1]`
Test Case 2	`[10, 20]`	`[20, 10]`

Explanation (TC 1): Swapping elements from ends towards center: swap(1,5) and swap(2,4) gives [5,4,3,2,1].

Problem 3: Check if Array is Sorted

Very Easy

Description: Write a function that checks whether a given array of integers is sorted in non-decreasing (ascending) order. Return `true` if sorted, otherwise return `false`.

Sample	Input Array	Expected Output
Test Case 1	`[10, 20, 30, 40, 50]`	`true`
Test Case 2	`[1, 5, 3, 9]`	`false`

Explanation (TC 2): Element 5 is followed by 3 (5 > 3), breaking ascending order.

Problem 4: Linear Search in Array

Very Easy

Description: Given an array `arr` and a target number K, find the first index where K occurs. If K is not present in the array, return `-1`.

Sample	Input (arr, K)	Expected Output
Test Case 1	`arr = [4, 2, 7, 1, 9], K = 7`	`2`
Test Case 2	`arr = [10, 20, 30], K = 50`	`-1`

Explanation (TC 1): Number 7 is found at index 2 (0-indexed: 4@0, 2@1, 7@2).

Problem 5: Count Even and Odd Numbers

Very Easy

Description: Given an array of integers, count the total number of even elements and odd elements present in it.

Sample	Input Array	Expected Output
Test Case 1	[1, 2, 3, 4, 5, 6]	Even: 3, Odd: 3
Test Case 2	[8, 12, 20]	Even: 3, Odd: 0

Explanation (TC 1): Evens are 2, 4, 6 (count=3); Odds are 1, 3, 5 (count=3).

Problem 6: Find Sum and Average of Elements

Very Easy

Description: Calculate the total sum of all elements in an integer array, then compute the average value (Sum / N).

Sample	Input Array	Expected Output
Test Case 1	[10, 20, 30, 40]	Sum: 100, Average: 25.0
Test Case 2	[5, 5, 5, 5, 5]	Sum: 25, Average: 5.0

Explanation (TC 1): Total Sum = 10+20+30+40 = 100. Average = 100 / 4 = 25.0.

Part 1: Basic Array Problems (Questions 7 – 10)

Problem 7: Move All Zeros to the End

Easy

Description: Given an array of integers, move all 0s to the end of the array while maintaining the relative order of all non-zero elements.

Sample	Input Array	Expected Output
Test Case 1	[0, 1, 0, 3, 12]	[1, 3, 12, 0, 0]
Test Case 2	[4, 5, 0, 0, 1]	[4, 5, 1, 0, 0]

Explanation (TC 1): Non-zero elements [1, 3, 12] stay in order, and two zeros are pushed to the end.

Problem 8: Find the Missing Number

Easy

Description: You are given an array containing N-1 distinct integers in the range from 1 to N. One number is missing. Find and return the missing number.

Sample	Input (arr, N)	Expected Output
Test Case 1	arr = [1, 2, 4, 5, 6], N = 6	3
Test Case 2	arr = [2, 3, 1, 5], N = 5	4

Explanation (TC 1): Expected sum of 1 to 6 is 21. Actual sum is 1+2+4+5+6 = 18. Missing = 21 - 18 = 3.

Problem 9: Remove Duplicates from Sorted Array

Easy

Description: Given a sorted array, remove the duplicate elements in-place such that each unique element appears only once. Return the unique elements array or length.

Sample	Input Array	Expected Output
Test Case 1	[1, 1, 2, 2, 3, 4]	[1, 2, 3, 4] (Length: 4)
Test Case 2	[0, 0, 0, 1]	[0, 1] (Length: 2)

Explanation (TC 1): Duplicates 1 and 2 are removed, leaving unique elements [1, 2, 3, 4].

Problem 10: Rotate Array Right by 1 Position

Easy

Description: Given an array, rotate its elements clockwise (to the right) by one position so that the last element becomes the first element.

Sample	Input Array	Expected Output
Test Case 1	[1, 2, 3, 4, 5]	[5, 1, 2, 3, 4]
Test Case 2	[9, 8, 7]	[7, 9, 8]

Explanation (TC 1): Last element 5 moves to front at index 0, pushing all other elements one step right.

Part 2: Basic String Problems (Questions 11 – 12)

Problem 11: Reverse a String

Very Easy

Description: Given a string S, write a program to reverse all characters in the string and return the newly formed string.

Sample	Input String	Expected Output
Test Case 1	"hello"	"olleh"
Test Case 2	"Java"	"avaJ"

Explanation (TC 1): Reversing characters of "hello" from end to start gives "olleh".

Problem 12: Check for Palindrome String

Very Easy

Description: Check whether a given string reads the same forwards and backwards. Return `true` if it is a palindrome, otherwise return `false`.

Sample	Input String	Expected Output
Test Case 1	"madam"	true
Test Case 2	"coding"	false

Explanation (TC 1): "madam" reversed is still "madam" (equal content).

Part 2: Basic String Problems (Questions 13 – 20)

Problem 13: Count Vowels and Consonants

Very Easy

Description: Given a string, count the total number of vowels ('a', 'e', 'i', 'o', 'u') and consonants present in it. Ignore spaces and numbers.

Sample	Input String	Expected Output
Test Case 1	"automation"	Vowels: 6, Consonants: 4
Test Case 2	"sky"	Vowels: 0, Consonants: 3

Explanation (TC 1): Vowels in "automation": a, u, o, a, i, o (6). Consonants: t, m, t, n (4).

Problem 14: Toggle Case of Characters

Easy

Description: Convert all lowercase letters in a string to uppercase, and all uppercase letters to lowercase.

Sample	Input String	Expected Output
Test Case 1	"JaVa"	"jAvA"
Test Case 2	"Hello World"	"hELLO wORLD"

Explanation (TC 1): Upper 'J'->'j', lower 'a'->'A', upper 'V'->'v', lower 'a'->'A'.

Problem 15: Find First Non-Repeating Character

Easy

Description: Given a string, find the first character that does not repeat anywhere else in the string. If all repeat, return `'-'`.

Sample	Input String	Expected Output
Test Case 1	`"swiss"`	`'w'`
Test Case 2	`"aabb"`	`'-'`

Explanation (TC 1): 's' repeats 3 times. 'w' appears only once and is the first non-repeating char.

Problem 16: Check if Two Strings are Anagrams

Easy

Description: Two strings are anagrams if they contain the exact same characters with the same frequencies, rearranged. Return `true` or `false`.

Sample	Input (s1, s2)	Expected Output
Test Case 1	`s1 = "listen", s2 = "silent"`	`true`
Test Case 2	`s1 = "hello", s2 = "billion"`	`false`

Explanation (TC 1): "listen" and "silent" both have letters {e, i, l, n, s, t} in frequency 1.

Problem 17: Remove All Spaces from String

Very Easy

Description: Given a sentence or string containing spaces, return a new string with all white spaces removed.

Sample	Input String	Expected Output
Test Case 1	`"Data Structures"`	`"DataStructures"`
Test Case 2	`" a b c "`	`"abc"`

Explanation (TC 1): Space between "Data" and "Structures" is removed.

Problem 18: Find Maximum Occurring Character

Easy

Description: Given a string, find the character that appears the highest number of times in the string.

Sample	Input String	Expected Output
Test Case 1	`"testsample"`	`'e' (or 't', count = 2)`
Test Case 2	`"apple"`	`'p' (count = 2)`

Explanation (TC 2): In "apple", character 'p' appears 2 times, higher than any other letter.

Problem 19: Longest Common Prefix

Easy

Description: Given an array of strings, find the longest common starting prefix among all strings. If none, return `""`.

Sample	Input Array of Strings	Expected Output
Test Case 1	`["flower", "flow", "flight"]`	`"fl"`
Test Case 2	`["dog", "racecar", "car"]`	`""`

Explanation (TC 1): "fl" is the longest starting prefix common to all three words.

Problem 20: Check String Rotation

Medium

Description: Given two strings `s1` and `s2`, check if `s2` is a valid circular rotation of `s1`. (Hint: check if `s2` is a substring of `s1 + s1`).

Sample	Input (s1, s2)	Expected Output
Test Case 1	`s1 = "waterbottle", s2 = "erbottlewat"`	`true`
Test Case 2	`s1 = "hello", s2 = "lloeh"`	`false`

Explanation (TC 1): Rotating "waterbottle" right by 2 positions yields "erbottlewat".